

FREESTANDING BENCH · 65-1/2 X 24-3/4 IN · 5 IN CLEAR FLOOR

Freestanding Bench Cut List

The client's change of 2026-10-26: the bench is now a freestanding piece with 5 in of clear floor under it for the robot vacuum. The alcove's baseboard stays: the top slides in 1/4 in off each side wall and 1/4 in off the back wall, and the posts stand inside it, the rear pair 2 in off the back wall, clear of the baseboard with no notch (Brian, 2026-11-01). Kept from the built-in: all-white frame-and-panel fronts with brass pulls, the solid maple top at 14 in (seat 17 in with the cushion), two drawers on Blum 563H undermounts (18 in now: standing the posts off the baseboard is what the case depth paid for), 1/2 in Baltic birch boxes, Osmo Natural on the bare maple.

The base is decided: on 2026-10-26 Brian picked A of five leg options drawn from the drawer bench's legs. It is built the way that bench is: four proud 2-1/2 in square soft maple corner posts, floor to the underside of the top, 3/8 in chamfers on all four vertical edges, the painted ends and the back housed in stopped grooves in the posts 1/2 in behind the post faces, the drawer fronts running between the front posts 1/4 in behind their faces. The earlier version on short legs under a box is superseded; the base options page records the pick. Drawers are 6-1/2 in tall instead of 8-1/2 (the 5 in of floor come out of the drawers); no plinth, no scribe strips, no wall screws. The built-in design stays on its own shop page in case the client goes back.

31 pieces: the case first, then the drawers, each grouped by material in cutting order. Inches to the nearest 1/32. Every part has a shop drawing with its grooves, rabbets, notches and bores located from

the blank's own edges.


GOOD
DOGS
LIVE HERE

NUMBERS THAT MOVED

- **Posts** 2-1/2 x 2-1/2 x 13-1/4 in soft maple, 3/8 in chamfers, outer faces 62-1/2 in apart; no notch, the rear posts stand 2 in off the back wall, 3/4 in clear of the baseboard and shoe
- **Clear floor under the case** 5 in (the tallest current Roomba is under 4 in); only the posts reach the floor
- **Top** 65-1/2 x 24-3/4 in, 1/4 in of air to each side wall, rear edge 1/4 in off the back wall; overhangs the posts 1-1/2 in at the ends, 3/4 in at the front and 1-3/4 in at the back (was 66 x 24-5/8, wall to wall)
- **Drawer fronts** 28-9/16 x 8-1/8 in between the front posts (were 31-3/32 x 10-1/8)
- **Drawer box** 27-11/16 x 6-1/2 x 18 in, 5-17/32 in inside (was 8-1/2 in tall)
- **Runner depth** 516.1 mm from the fronts' back faces to the back's inside face against Blum's 481 for the 18 in runner; on the post side the runner tail sits 44.6 mm in front of the back for the rear bracket (Blum allows 6 to 48; 471 mm runner body, measured on the drawer bench)
- **Overall** 65-1/2 x 24-3/4 x 17 in with the cushion; wood top at 14 in, seat 17 in, unchanged

Measure the alcove at the floor, 16 in and 36 in, at the back wall and the pilaster faces, before cutting. The piece is 65-1/2 in wide; if the opening is under 66 in anywhere, narrow the top. The baseboard stays: the rear posts stand 2 in off the back wall with 3/4 in to spare over a 1-1/4 in baseboard and shoe, so measure how far they stand off the wall; the posts have no notch. The low receptacle seen in the photo ends up behind the case; nothing can stay plugged into it. Scrap-test the Blum hook bore in the 1/2 drawer backs (a 2 mm skin).

Case 11 pieces

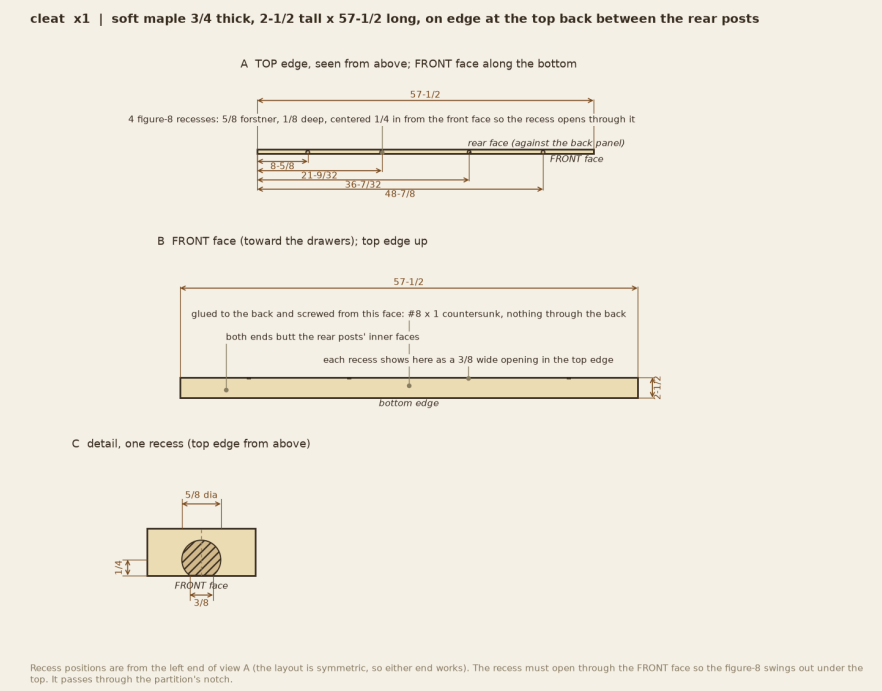
Soft maple: posts and the rear cleat (posts bare, Osmo Natural)

5 pieces · 1.9 sq ft face, no kerf or waste

1× cleat

3/4 x 2-1/2 x 57-1/2

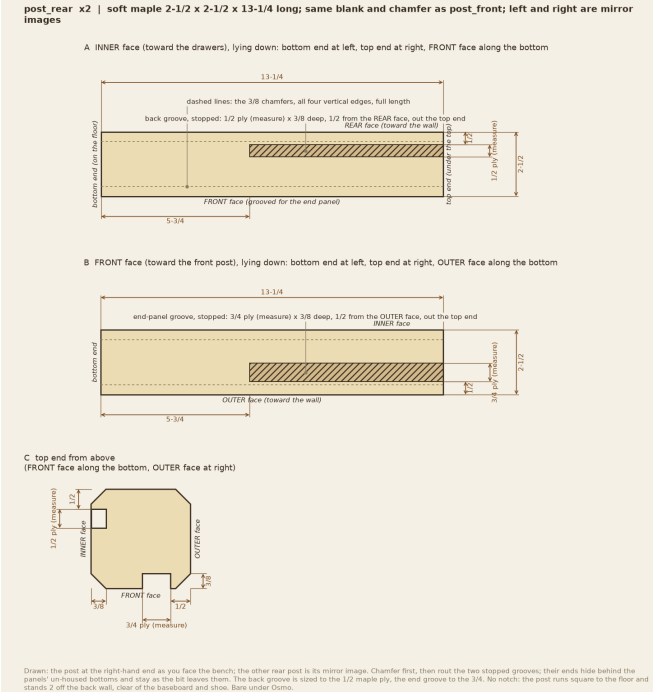
on edge against the back's inside face at the top, between the rear posts, through the partition notch; four 5/8 figure-8 recesses 1/8 deep on the top edge, centered 1/4 in from the front face; glued and screwed to the back from inside (#8 x 1 countersunk, nothing through the back)



2× post_rear see drawing: not a plain rectangle

2-1/2 x 2-1/2 x 13-1/4

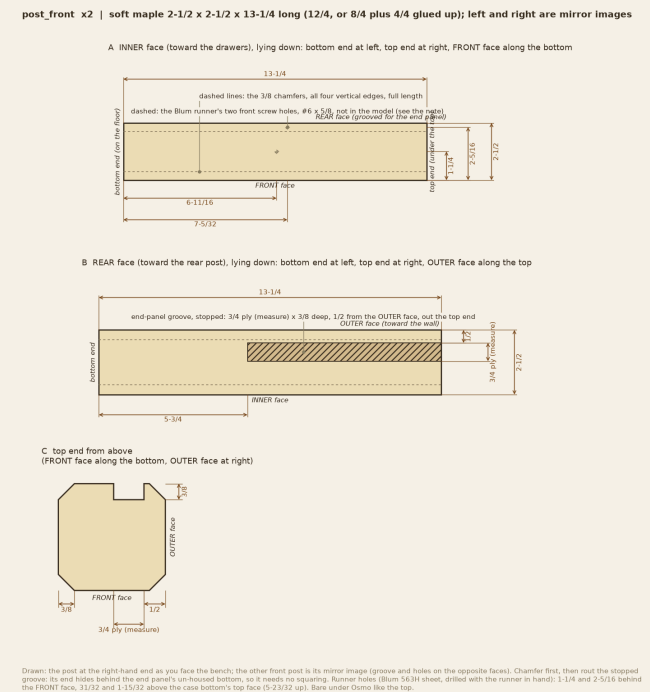
2-1/2 square x 13-1/4 (12/4 stock or an 8/4 plus 4/4 glue-up); 3/8 chamfer on all four vertical edges, full length, cut before the grooves; end-panel groove 23/32 wide (measure) x 3/8 deep on the face toward the other post of that end, 1/2 in from the outer face, from 5-3/4 up out the top (the end stays as the bit leaves it, the panel's notched bottom covers it); bare under Osmo; left/right mirrored; plus the back's groove 15/32 wide (measure) x 3/8 deep on the inner face, 1/2 in from the rear face, same stop; no notch, the post stands 3/4 clear of the baseboard



2× post_front see drawing: not a plain rectangle

2-1/2 x 2-1/2 x 13-1/4

2-1/2 square x 13-1/4 (12/4 stock or an 8/4 plus 4/4 glue-up); 3/8 chamfer on all four vertical edges, full length, cut before the grooves; end-panel groove 23/32 wide (measure) x 3/8 deep on the face toward the other post of that end, 1/2 in from the outer face, from 5-3/4 up out the top (the end stays as the bit leaves it, the panel's notched bottom covers it); bare under Osmo; left/right mirrored; the Blum runner front screws into the inner face, holes 1-1/4 and 2-5/16 behind the front face, #6 x 5/8



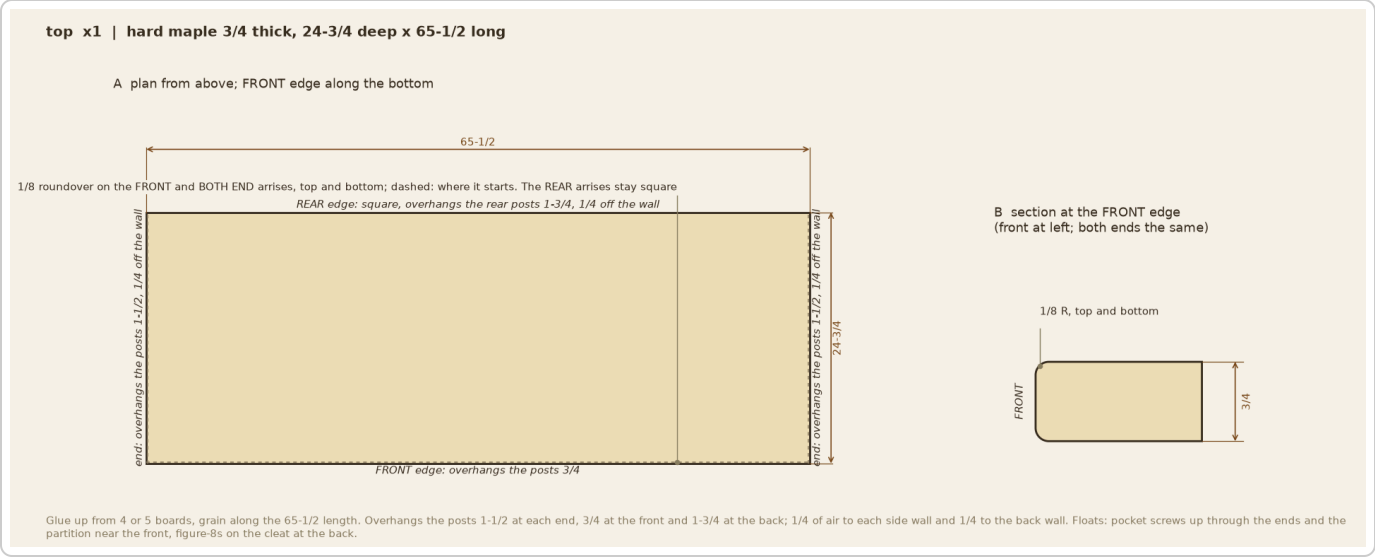
Hard maple: top (bare, Osmo Natural)

1 pieces · 11.3 sq ft face, no kerf or waste

1× top

3/4 x 24-3/4 x 65-1/2

glue up from 4 or 5 boards, grain along the 65-1/2; 1/8 roundover on the front and both end arrises, top and bottom (the back edge stays square against the wall); overhangs the posts 1-1/2 at each end, 3/4 at the front and 1-3/4 at the back; pocket screws up through the ends and partition near the front, figure-8s on the cleat at the back



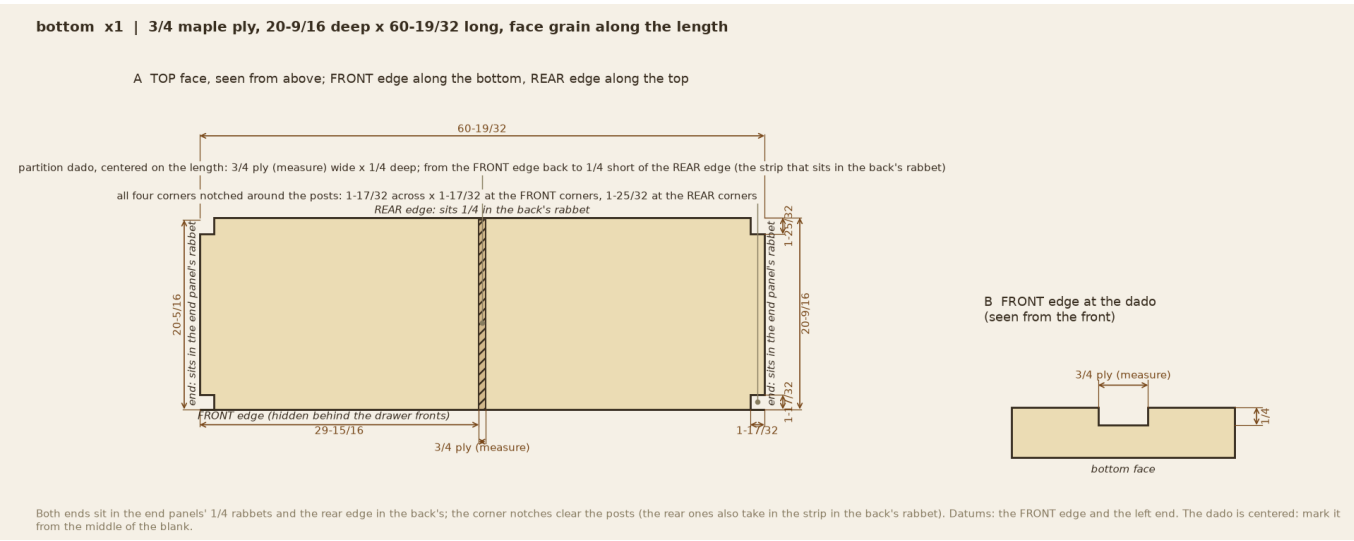
3/4 in maple plywood (measure the sheet): ends (show faces, painted white, bottom edges banded), bottom, partition

4 pieces · 11.8 sq ft face, no kerf or waste

1× bottom

23/32 x 20-9/16 x 60-19/32

face grain across the 60-19/32; sits 1/4 into the rabbets of both ends and the back; all four corners notched 1-17/32 across x 1-17/32 (front) and 1-25/32 (rear) around the posts; 1/4 deep x 3/4 ply (measure) dado for the partition, centered, from the back to the front edge; front edge covered by the drawer fronts



1× partition

23/32 x 7-25/32 x 20-5/16

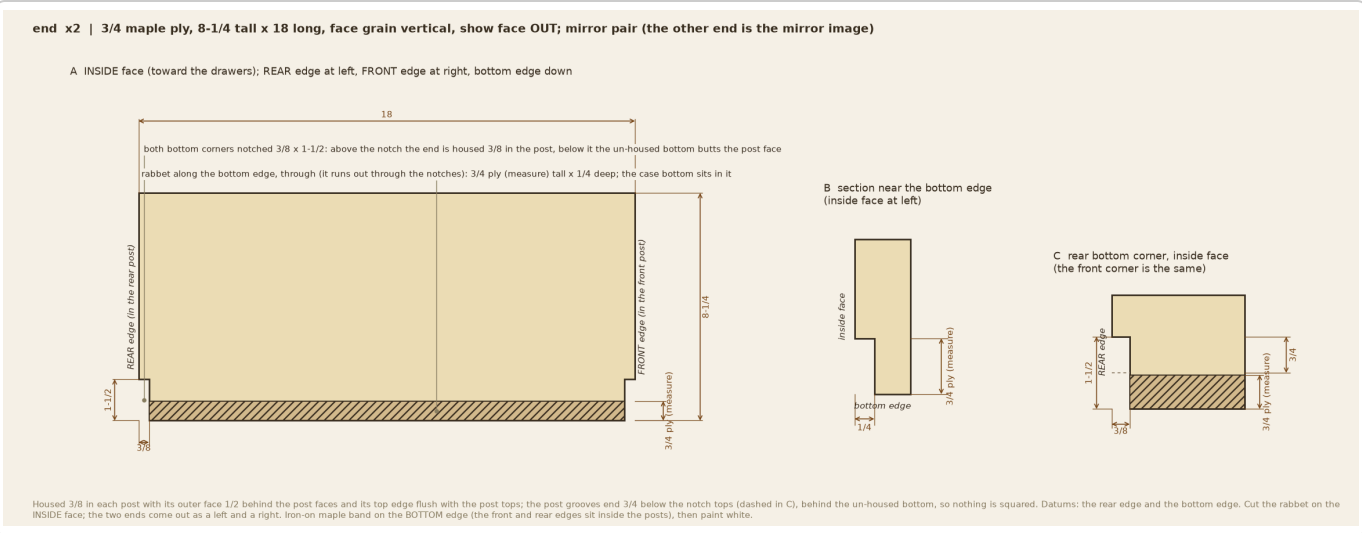
face grain vertical; sits in the bottom dado and against the back; front edge on the fronts' back faces (the partition-side runners mount to it the frameless way); notch the top rear corner 3/4 deep x 2-1/2 tall for the cleat



2× end see drawing: not a plain rectangle

23/32 x 8-1/4 x 18

show face out, face grain vertical, painted white (bottom edge banded); housed 3/8 in each post, outer face 1/2 behind the post faces, top edge flush with the post tops; both bottom corners notched 3/8 (from each end) x 1-1/2 tall: the housed tongue starts 3/4 above the post groove's end and the panel's bottom 1-1/2 butts the post faces; 1/4 x 23/32 (measure) rabbet along the bottom inside edge for the case bottom, through; mirror pair



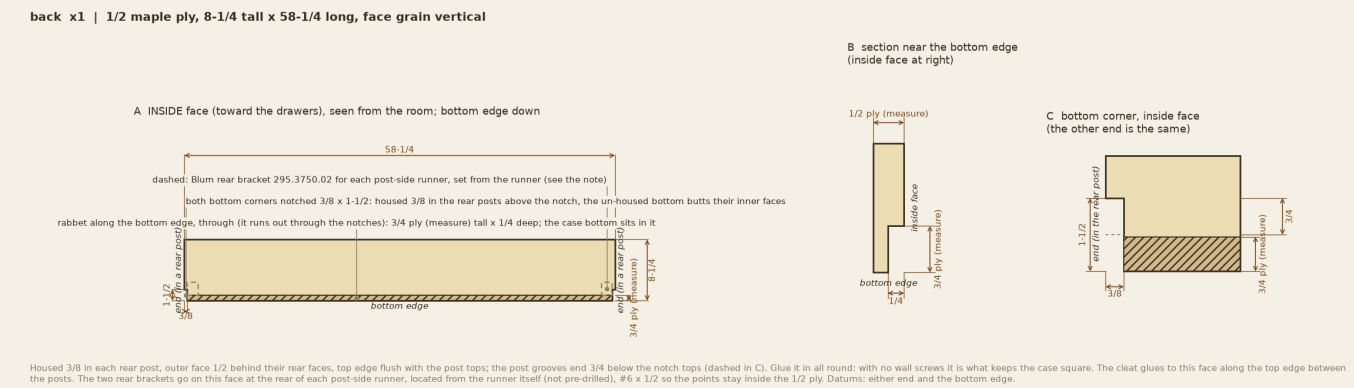
1/2 in maple plywood (measure the sheet): back, housed in the rear posts

1 pieces · 3.3 sq ft face, no kerf or waste

1× **back** see drawing: not a plain rectangle

15/32 x 8-1/4 x 58-1/4

housed 3/8 in each rear post, outer face 1/2 behind their rear faces, top edge flush with the post tops; both bottom corners notched 3/8 (from each end) x 1-1/2 tall: the housed tongue starts 3/4 above the post groove's end and the panel's bottom 1-1/2 butts the post faces; 1/4 x 23/32 (measure) rabbet along the bottom inside edge for the case bottom, through; glued in (with no wall screws it is what keeps the case square); the Blum rear brackets screw to its inside face with #6 x 1/2



Drawers 20 pieces

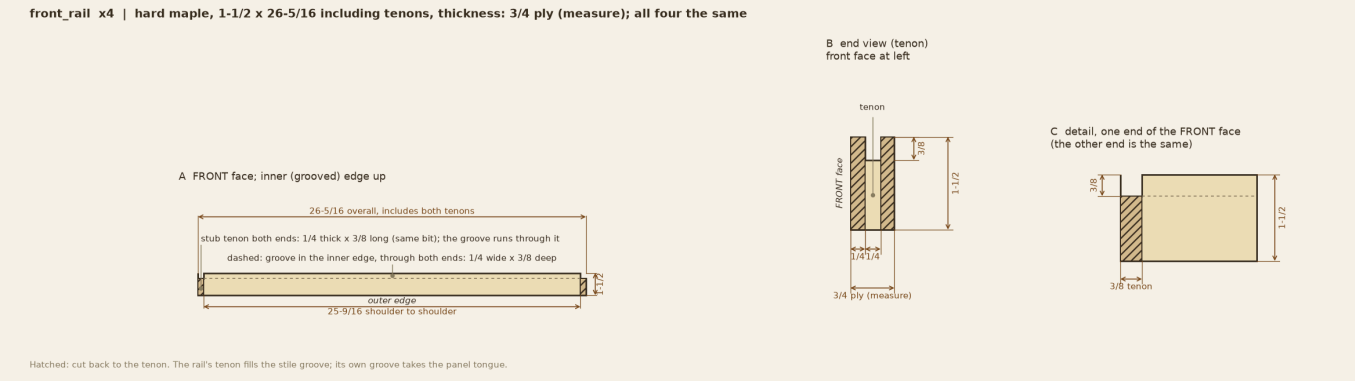
Drawer front frames, hard maple, painted white: stiles and rails

8 pieces · 1.4 sq ft face, no kerf or waste

4× front_rail see drawing: not a plain rectangle

23/32 x 1-1/2 x 26-5/16

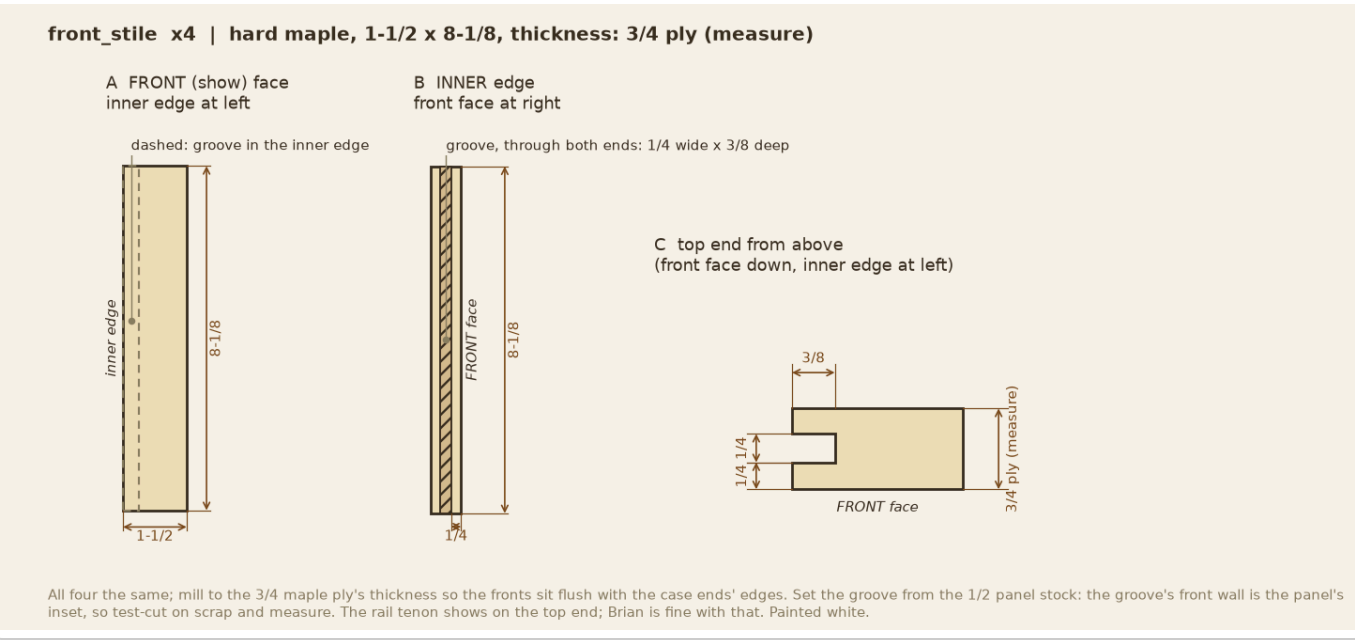
1/4 x 3/8 groove on the inner edge, through; 1/4 x 3/8 stub tenon on both ends (same bit); length includes both tenons; painted white



4× front_stile see drawing: not a plain rectangle

23/32 x 1-1/2 x 8-1/8

1/4 x 3/8 groove on the inner edge, through, front wall 1/4 thick (set the fence from the panel ply); the rail tenons show on both ends (the top one is the visible one); painted white



Drawer front panels, 1/2 in MDF or paint-grade plywood (measure the sheet), painted white with the frames

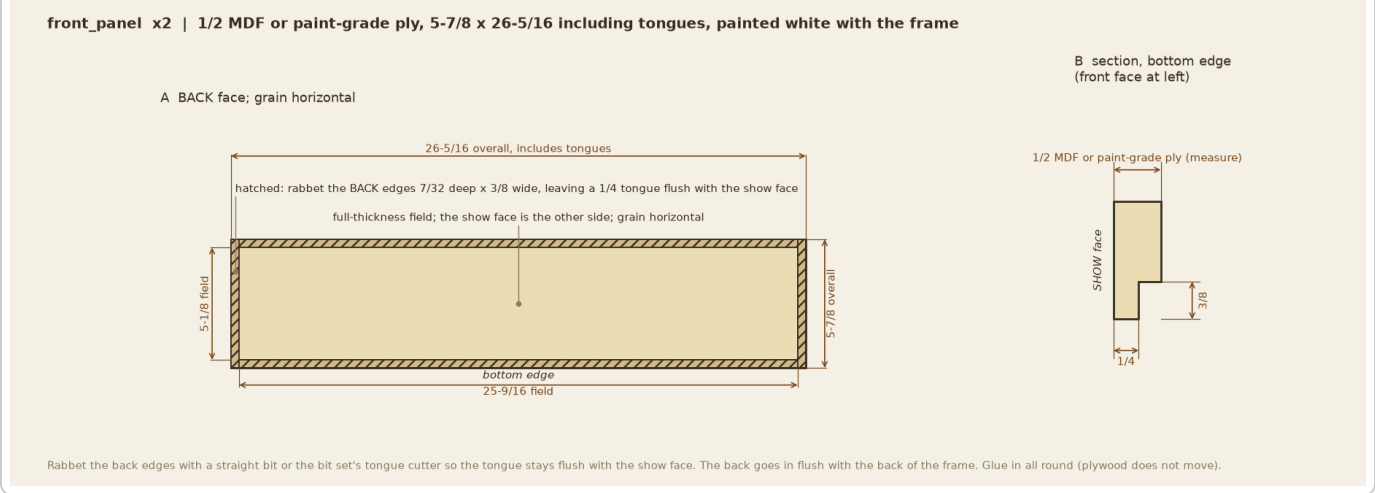
2 pieces · 2.1 sq ft face, no kerf or waste

2× front_panel

see drawing: not a plain rectangle

15/32 x 5-7/8 x 26-5/16

1/4 x 3/8 tongue on all four edges, flush with the show face (rabbet the BACK edges 7/32 deep); grain horizontal; size includes the tongues; back flush with the frame back, face 1/4 below the maple; glue in (plywood, no movement)



Drawer boxes: sides, fronts, backs and bottoms, 1/2 in 9-ply Baltic birch (measure the sheet)

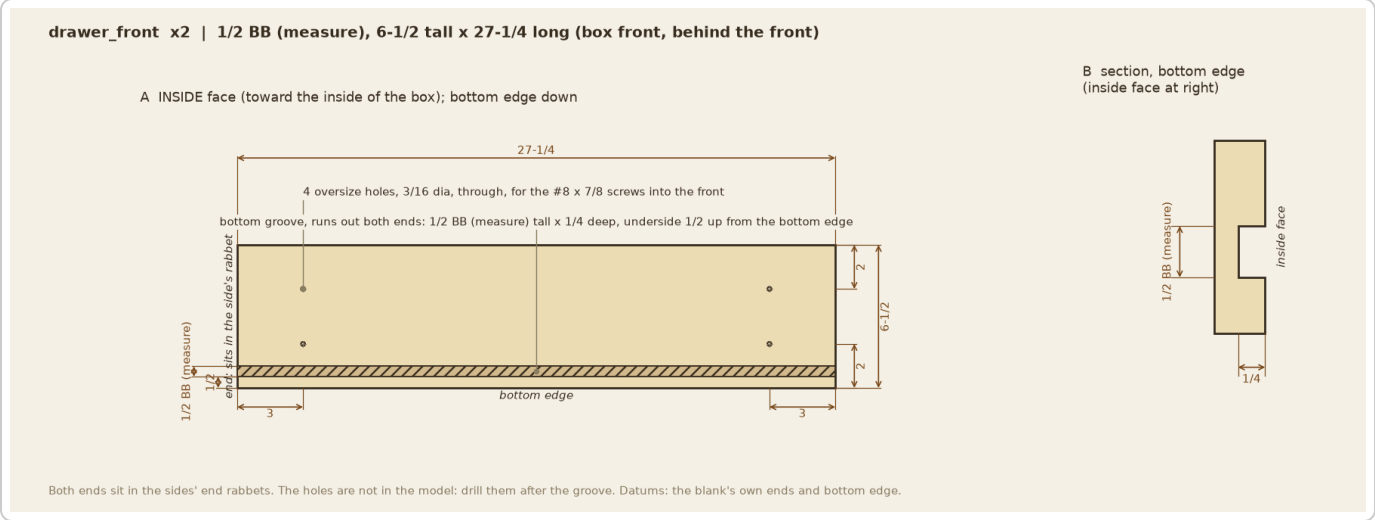
10 pieces · 14.7 sq ft face, no kerf or waste

2× drawer_front

see drawing: not a plain rectangle

15/32 x 6-1/2 x 27-1/4

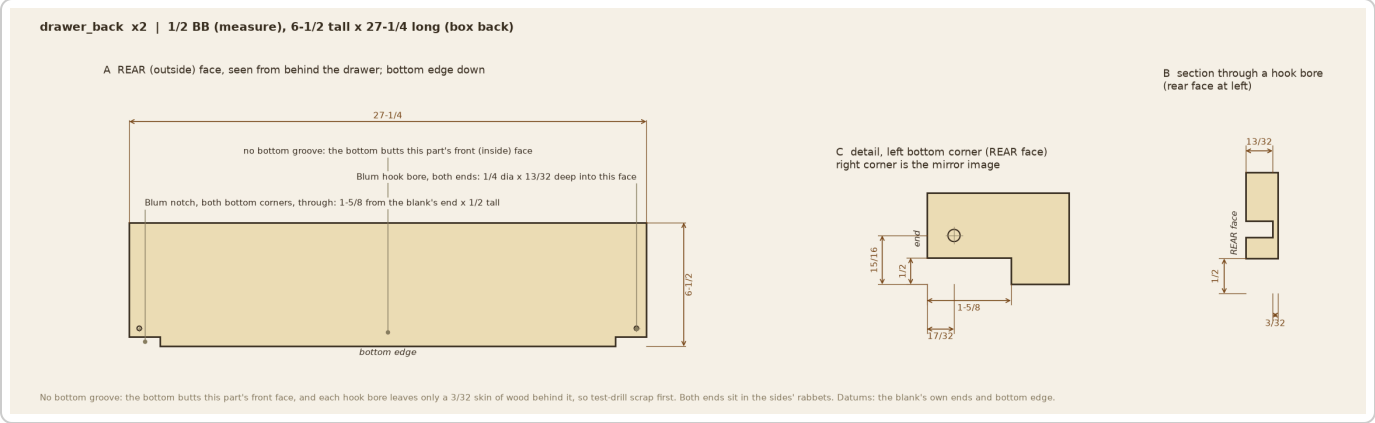
box front, sits in the side rabbets; 1/4 deep bottom groove to match the sides; the front screws to it from inside through 4 oversize holes: #8 x 7/8 in, pre-drilled, so the bite stays short of the face; the box front overlaps the top rail by 1-1/8 (best bite) and the panel field, but the bottom rail by only 1/4



2× drawer_back

15/32 x 6-1/2 x 27-1/4

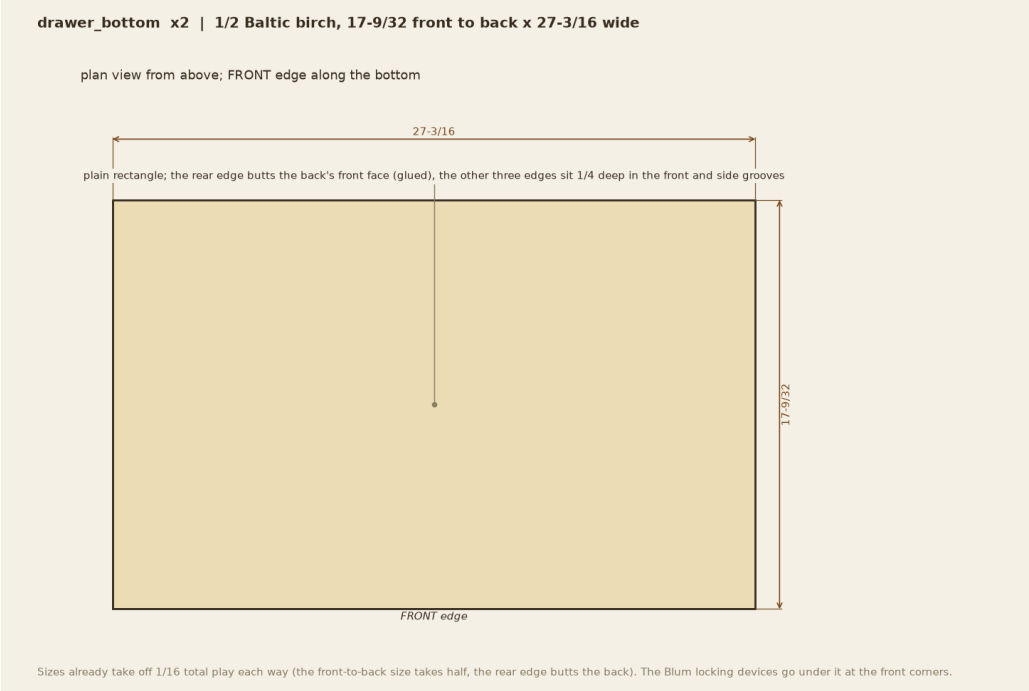
box back, sits in the side rabbets; no bottom groove (a groove would break into the hook bores), the bottom butts its front face; Blum notch 1-3/8 in from each side's inner face x 1/2 tall at both bottom corners; hook bores 1/4 dia x 13/32 deep into the rear face, centered 9/32 in from each side's inner face and 15/16 above the bottom edge (Blum 563H sheet)



2× drawer_bottom

15/32 x 17-9/32 x 27-3/16

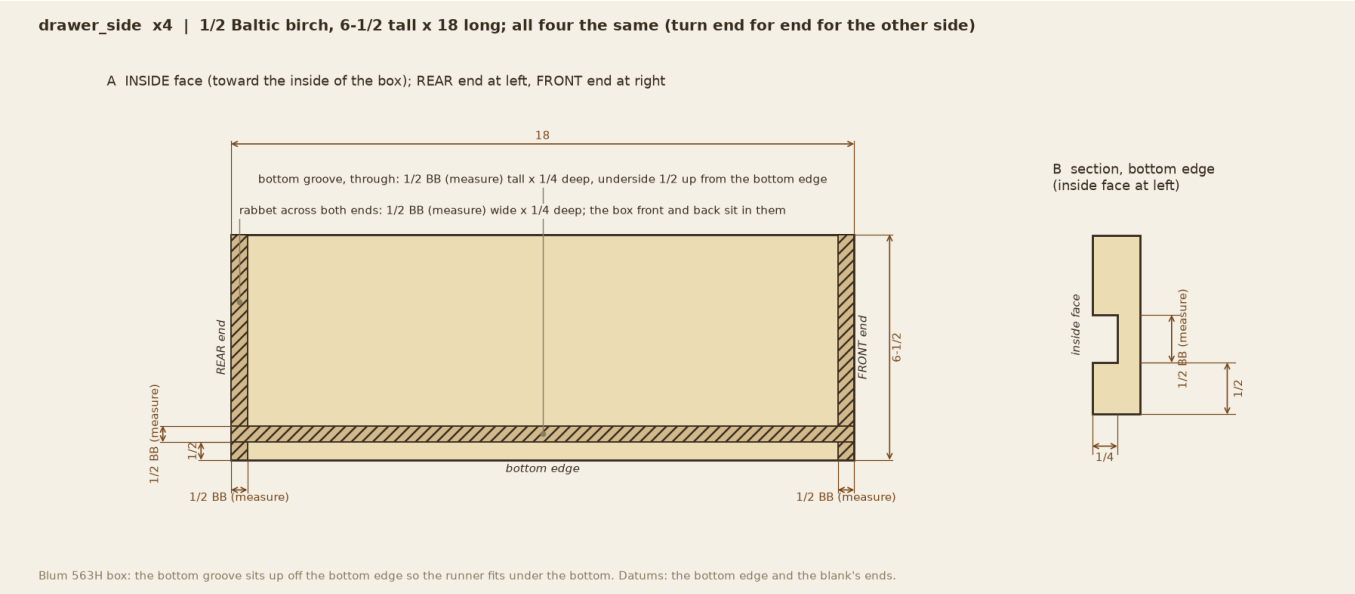
plain rectangle: 1/4 into the side and front grooves, butted to the back and glued to it; the Blum locking devices sit flush under it at the front corners (their screws go into the box front)



4× drawer_side see drawing: not a plain rectangle

15/32 x 6-1/2 x 18

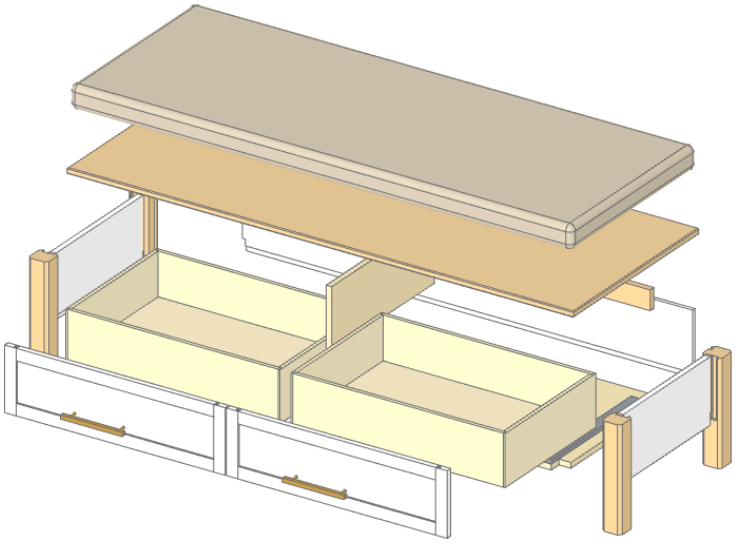
1/4 deep x 1/2 BB (measure) rabbet across both ends on the inside face; 1/4 deep x 1/2 BB (measure) bottom groove, underside 1/2 up from the bottom edge, through



Hardware from the design; not in the cut list

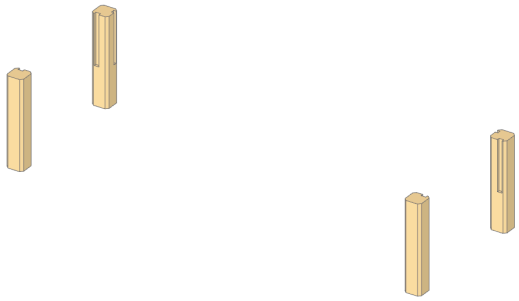
2 pair	**Blum TANDEM plus BLUMOTION 563H, 18 in (563H4570B)** one pair per drawer; 90 lb dynamic; runner fronts 1/16 behind the fronts' back faces
2	**Blum rear mounting brackets 295.3750.02** post-side runners: the runner front screws into the post's inner face (#6 x 5/8), the rear sits on a bracket screwed to the 1/2 back with #6 x 1/2; the partition-side runners mount to the partition as in a frameless case
2 pair	**Blum locking devices T51.1901 R and L** front corners under the drawer bottom, screwed into the 1/2 box front with #6 x 1/2 so the points stay inside it
2	**Brass bar pull, 8 in** centered on each front; break-away machine screws through the 1/2 panel plus the 1/2 box front
4	**Figure-8 tabletop fasteners** cleat to top; recess centered 1/4 in from the front face so it opens through
6	**Pocket screws #8 x 1-1/4 fine** ends and partition to the top, near the front
6	**Wood screws #8 x 1** cleat to the back from inside, countersunk; bottom to the partition from below
8	**Wood screws #8 x 7/8** fronts to the box fronts, through oversize holes, pre-drilled
1 roll	**Iron-on maple edge banding, 13/16** the ends' bottom edges, about 36 in all, painted with them
1	**Bench cushion** about 65 x 24-3/4 x 3, boxed, the full depth of the top

Assembly 10 steps, in order



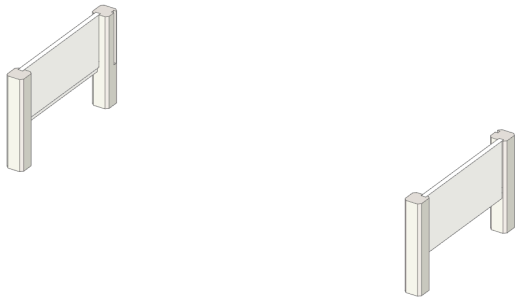
1. Posts

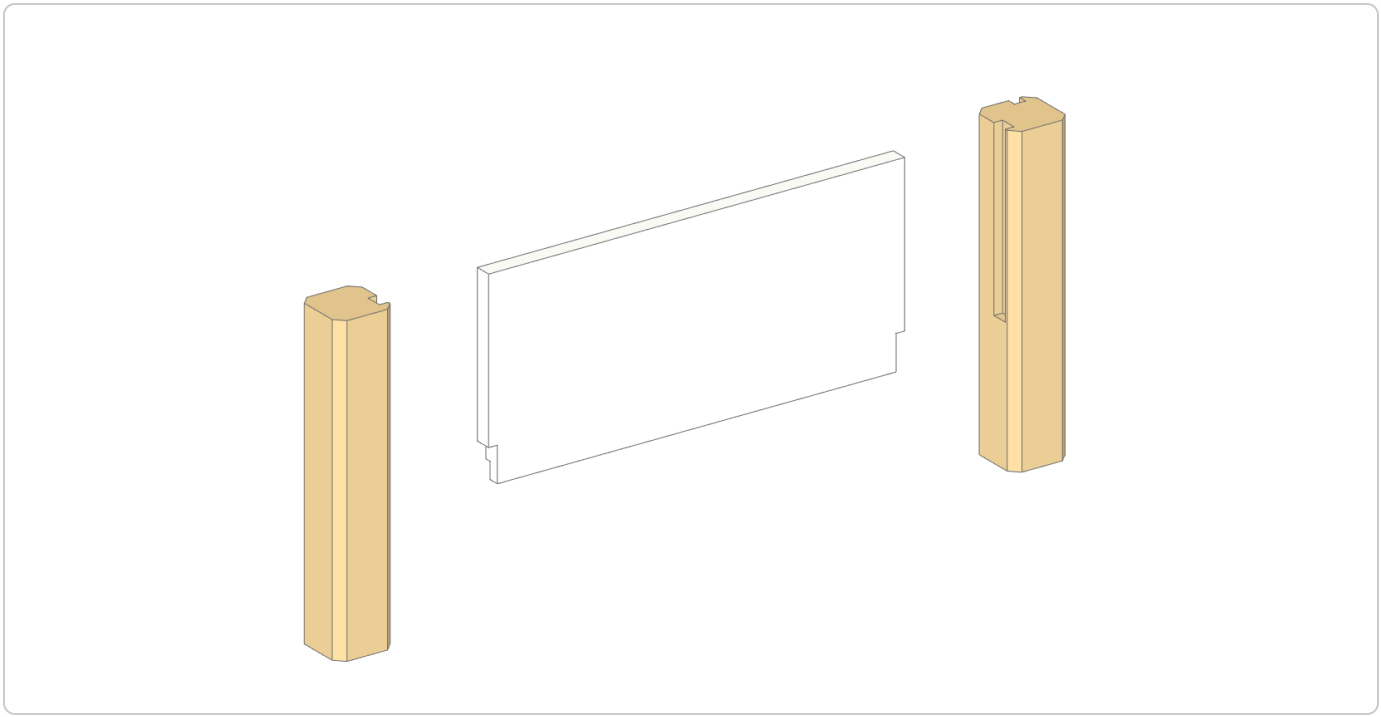
Four 2-1/2 square soft maple posts, 13-1/4 long. Chamfer the four vertical edges 3/8 first, full length, then cut the stopped grooves: the end-panel groove on the face toward the other post of that end, 1/2 in from the outer face, 3/8 deep, from 5-3/4 up and out the top; the rear posts also take the back's groove on the inner face, 1/2 in from the rear face, same stop. No notch: the rear posts stand 2 off the back wall, 3/4 clear of the baseboard and shoe. Drill the front posts' Blum runner screw holes on the inner face, 1-1/4 and 2-5/16 behind the front face.



2. End assemblies

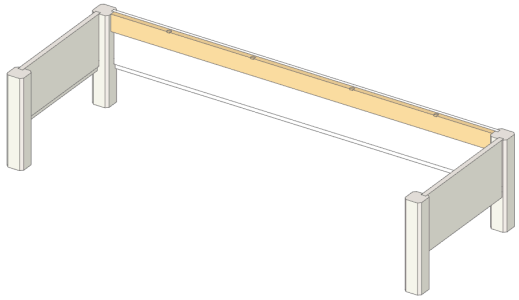
Glue each end panel into its front and rear post: tongues in the grooves above the 1-1/2 notches, the notched bottom on the post faces, top edge flush with the post tops, show face out. Check that each end is square and both posts stand plumb before the glue sets.





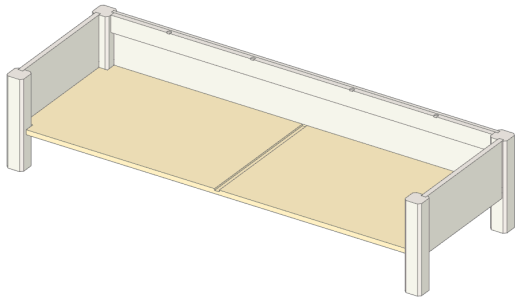
3. Back and cleat

Slide the 1/2 back down the rear posts' grooves to join the two end assemblies, notched bottom on the post faces, and glue it: with no wall screws it is what keeps the case square. Glue the cleat inside along the back's top edge between the rear posts and screw it from the inside (#8 x 1, countersunk, nothing through the back).



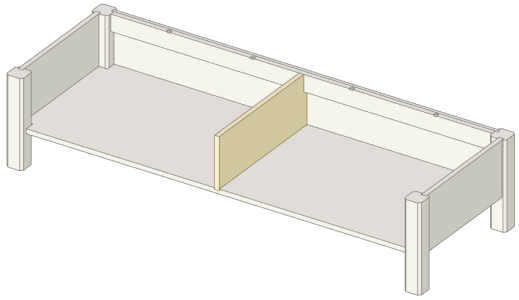
4. Bottom

Slide the case bottom in from the front into the 1/4 rabbets of both ends and the back, its corner notches around the four posts, and glue it. Its underside is the ceiling of the 5 in of clear floor; screw up through it into the partition after the next step.



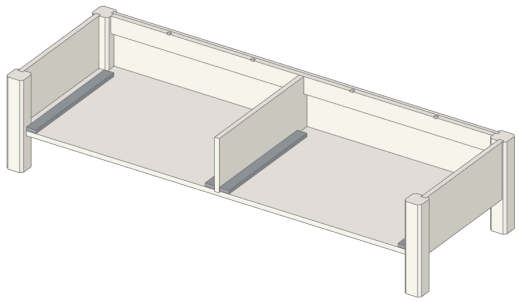
5. Partition

Drop the partition into the dado in the bottom, against the back, with its notch at the top rear around the cleat and its front edge on the fronts' back-face line, 31/32 behind the post faces. Screw up through the bottom.



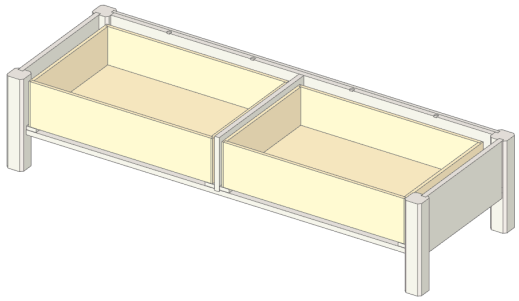
6. Slides and brackets

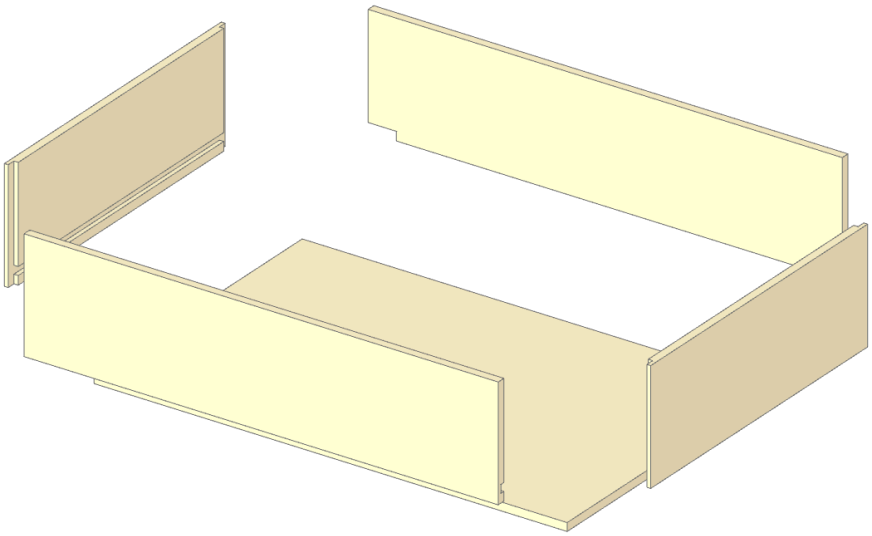
Post-side runners: front screws into the posts' inner faces (#6 x 5/8), rears on Blum brackets screwed to the back with #6 x 1/2. Partition-side runners: screwed to the partition the frameless way. All four level on the case bottom, fronts 1/16 behind the fronts' back-face line.



7. Drawer boxes

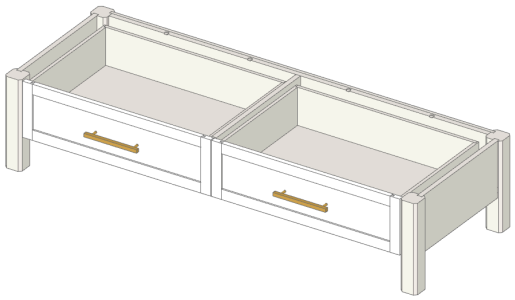
Glue each 1/2 birch box up around its bottom, 6-1/2 in tall: front and back in the side rabbets, bottom in the side and front grooves with its underside 1/2 above the side edges, butted to the back and glued to it. Notches and hook bores at the rear (drill scrap first: the bore leaves a thin skin). Clip the locking devices under the bottom at the front corners with #6 x 1/2 screws and set the boxes on the runners.

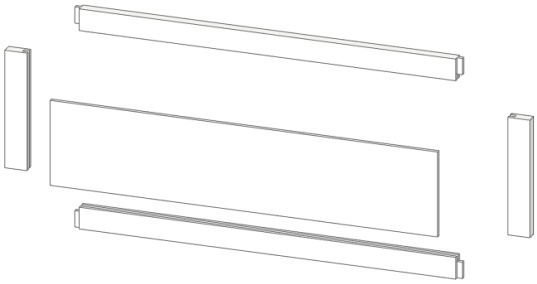




8. Fronts and pulls

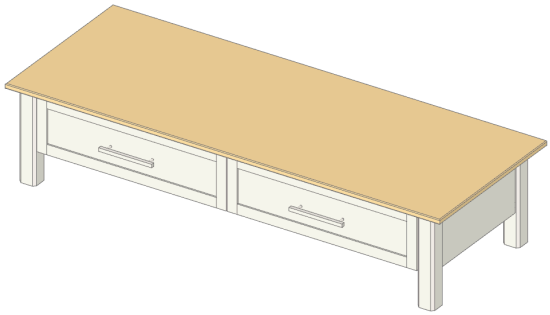
Glue up each 1/2 panel in its maple frame, back faces flush so the panel sits 1/4 below the maple (groove every piece first, then cut the tongues and tenons to fit). Prime and paint the fronts and the ends white. Hang the fronts between the front posts: 1/8 to each post, 1/8 between them, 1/8 under the top, bottoms flush with the case underside, faces 1/4 behind the post faces, screwed from inside the box through oversize holes. Center the pulls.





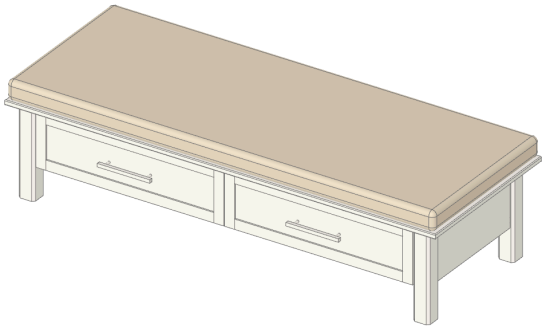
9. Top

Set the 65-1/2 x 24-3/4 top overhanging the posts 1-1/2 at each end, 3/4 at the front and 1-3/4 at the back. Pocket screws up into it through the ends and partition near the front; figure-8s on the cleat at the back so it can move. Osmo Natural on the top and the posts.



10. Cushion

Cushion on. Slide the bench into the alcove with the top 1/4 off the back wall and 1/4 off each side wall, the rear posts clear of the baseboard. Done.



Generated 2026-11-01 from `projects/Freestanding-bench/cutlist.csv` and `images/parts/`, which `freestanding_bench.py` and `part_drawings.py` write on every export; every drawn cut is probed against the CAD solid before its sheet is written. The bench image at the top and the assembly pictures are screenshots of the CAD viewer (`hero_shot.py`, `assembly_shots.py`). Part names match the CAD viewer. The built-in version of this bench is at its own page.